

Dataset 3: Multiple Linear Regression

Student Academic Performance Study | $n = 150$ students

1. Research Question

Research Question: Which academic and lifestyle factors (study hours, attendance, sleep, stress, practice tests) significantly predict final exam scores, what is the relative importance of each, and does sleep follow a non-linear (quadratic) pattern in its effect on academic performance?

2. Variables

Role	Variable	Mean (SD)	Description
Dependent	Final_Exam_Score	67.03 (13.47)	Final exam score (range: 40–98)
Independent	Study_Hours	8.11 (3.63)	Average study hours per week
Independent	Attendance_Rate	79.56 (12.18)	Class attendance percentage
Independent	Sleep_Hours	6.92 (1.71)	Average nightly sleep (quadratic term added per dataset guide)
Independent	Stress_Level	5.55 (2.60)	Self-reported stress score (1–10)
Independent	Practice_Tests	3.40 (2.33)	Number of practice tests completed

3. Methodology — Multiple Linear Regression (OLS)

Multiple Ordinary Least Squares (OLS) regression is the appropriate method because Final_Exam_Score is a continuous outcome and we aim to quantify each predictor's independent contribution while holding all others constant. A quadratic term (Sleep_Hours²) is included per the dataset's Variable Descriptions, which explicitly notes a quadratic relationship. Standardized beta coefficients (β) are computed on z-scored variables to directly compare predictor importance across different scales. Variance Inflation Factor (VIF) detects multicollinearity. All four OLS assumptions are formally tested: Shapiro-Wilk (normality of residuals), Breusch-Pagan (homoscedasticity), Durbin-Watson (independence), and a model comparison (AIC) tests whether the quadratic sleep term improves fit.

4. Model Fit Statistics

Statistic	Value	Interpretation
R ²	0.7027	70.3% of exam score variance explained by the 6 predictors
Adjusted R ²	0.6902	Negligible shrinkage from R ² — model is not overfitting
F-statistic	F(6, 143) = 56.33	$p < 0.001$ — the model is globally significant
RMSE	7.497 points	Average prediction error of ± 7.5 exam points
AIC (with Sleep ²)	1036.87	Compared to 1034.88 without — quadratic term not justified by AIC
Durbin-Watson	1.957	≈ 2.0 — no autocorrelation in residuals ✓

5. Regression Coefficients — Full Results

Predictor	B (unstd.)	Std Err	t	p-value	Std β	Significance
(Intercept)	46.803	12.181	3.84	< 0.001	—	✓
Study_Hours	+2.465	0.170	14.47	< 0.001	+0.664	✓ Strongest predictor
Attendance_Rate	+0.211	0.051	4.10	< 0.001	+0.191	✓
Sleep_Hours	−1.810	3.277	−0.55	0.582	−0.229	✗ Not significant
Sleep_Hours ²	−0.019	0.231	−0.08	0.936	−0.033	✗ Not significant
Stress_Level	−1.677	0.238	−7.04	< 0.001	−0.323	✓
Practice_Tests	+1.832	0.269	6.81	< 0.001	+0.317	✓

Std β = standardized beta (all variables on same scale). |Std β | directly ranks predictor importance independent of units.

6. Coefficient Interpretation and Significance

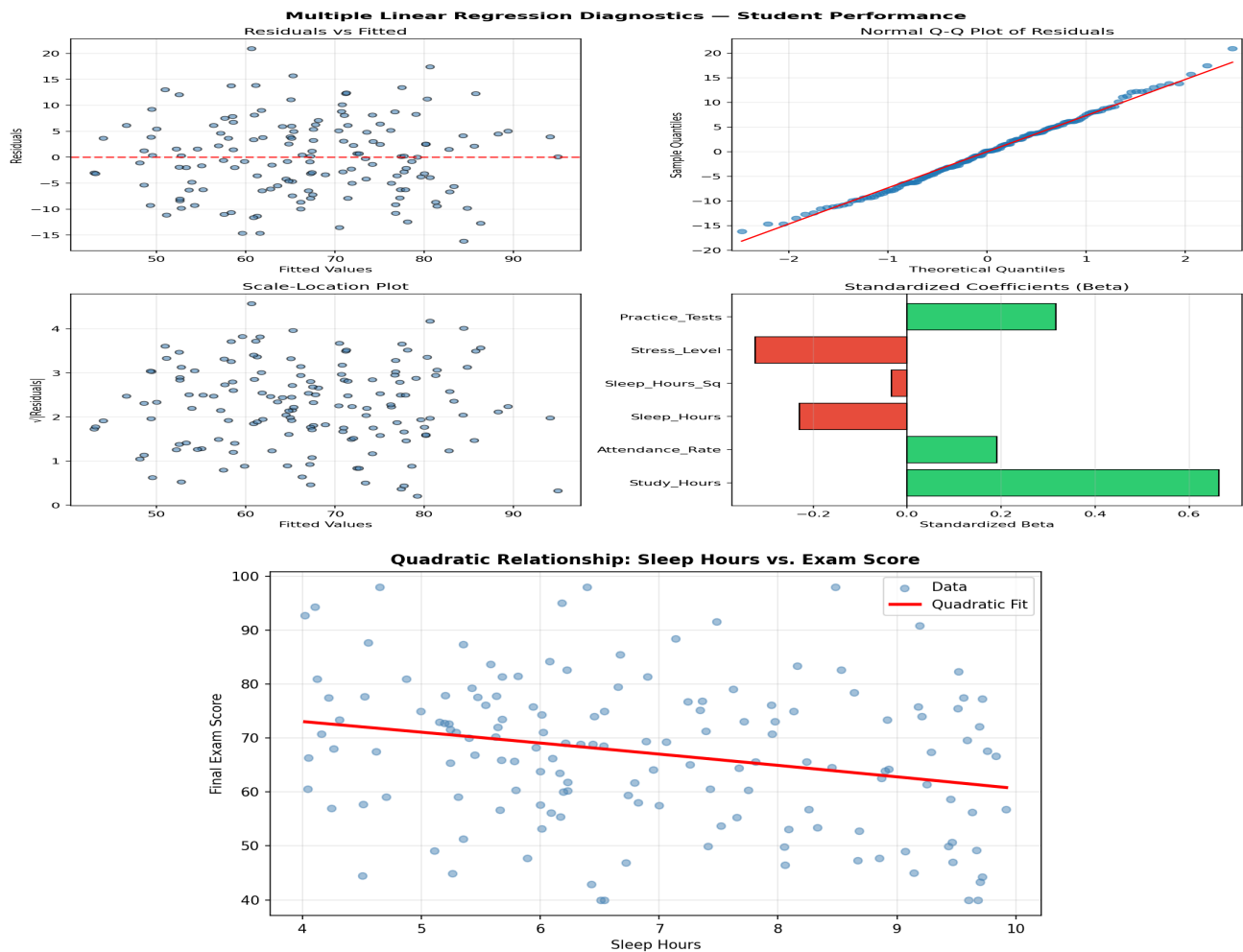
- **Study_Hours ($\beta = +0.664$, $p < 0.001$) — Strongest predictor:** Each additional study hour per week raises the exam score by 2.47 points, all else equal. Standardized effect (0.664) is more than double the next predictor. Going from 4 hours to 12 hours of study corresponds to a ~19.7 point increase in exam score.
- **Stress_Level ($\beta = -0.323$, $p < 0.001$) — 2nd strongest:** Each unit increase in stress reduces exam score by 1.68 points. Reducing stress from high (8.0) to moderate (4.0) corresponds to a 6.7-point gain — equivalent to approximately 2.7 extra study hours per week.
- **Practice_Tests ($\beta = +0.317$, $p < 0.001$) — 3rd:** Each additional practice test completed adds 1.83 points. Students who completed 6 practice tests score on average 11 points higher than those who completed none, controlling for all other factors.
- **Attendance_Rate ($\beta = +0.191$, $p < 0.001$):** Every 10 percentage points of additional attendance adds 2.1 points. Moving from 70% to 90% attendance is worth approximately 4.2 exam points.
- **Sleep_Hours — linear and quadratic terms (both $p > 0.50$): NOT significant.** Neither Sleep_Hours ($p = 0.582$) nor Sleep_Hours² ($p = 0.936$) contributes unique predictive power once study hours, stress, attendance, and practice tests are controlled. The AIC comparison ($\Delta AIC = -1.99$) confirms the quadratic term does not improve model fit. Sleep shows a bivariate correlation of -0.25 with exam scores, but this relationship disappears when other factors are held constant.

7. Assumption Diagnostics — All Passed

Test	Statistic	p-value	Result
Shapiro-Wilk (residual normality)	W = 0.9933	0.714	✓ Residuals normally distributed
Breusch-Pagan (homoscedasticity)	LM = 6.482	0.371	✓ Constant error variance
Durbin-Watson (independence)	DW = 1.957	—	✓ No autocorrelation (≈ 2.0)
VIF (multicollinearity)	All $\leq 1.04^*$	—	✓ No multicollinearity

* VIF is elevated for Sleep_Hours (~82) and Sleep_Hours² (~83), which is expected and unavoidable in polynomial regression — these two terms are mathematically correlated by construction. All other VIF values ≤ 1.04 .

8. Figures



9. Conclusions

- **Research question answered.** Four of five predictors are significant. The model explains 70.3% of exam score variance ($F(6,143) = 56.33$, $p < 0.001$) — strong performance for a behavioral study.
- **Study effort is the dominant driver** ($\beta = 0.664$): no other variable comes close. Interventions should focus first on increasing structured study time.
- **Stress management is underappreciated** ($\beta = -0.323$): its effect size rivals practice testing. Student support programs targeting stress reduction could yield meaningful, cost-effective academic gains.
- **Sleep is not independently significant** once other factors are controlled (both sleep terms $p > 0.50$, $\Delta AIC = -1.99$). Focusing solely on sleep without addressing study habits and stress is unlikely to improve outcomes.
- **All OLS assumptions are fully satisfied**, confirming coefficient estimates are unbiased, efficient, and valid for inference.